José Dias Neto - CV

Address	José Dias Neto Delft The Netherlands	Mobile Phone Personal Page Email	+31 6 28 71 97 11 **jdiasn.github.io** jdiasn@gmail.com

Professional Experience and Education

2025- **Data Manager** - Delft University of Technology, Netherlands
Management of the Ruisdael data flow from field instruments to public repositories

2021-2024 **PostDoc** - Delft University of Technology, Netherlands
Using the synergy of the cloud radars and lidar to create continuous wind profiles for studying turbulence in the boundary and cloud layers
Advisor: Dr. Louise Nuijens

2017-2021 **PhD in Meteorology** - University of Cologne, Germany
PhD's Thesis: Investigating aggregation in ice and snow clouds using novel combination of triple-frequency cloud radars and radar Doppler spectra
Advisor: Dr. Stefan Kneifel

2014-2017 **Research Assistant** - National Institute For Space Research, Brazil
Development of computational tools to read and evaluate large data arrays of Aerosol Optical Depth retrieved by new Generation of NOAA Polar Satellite System (SUOMI - NPP) over Brazilian territory
Advisor: Dr. Simone Sievert da Costa Coelho

2012-2014 **MSc in Meteorology** - National Institute For Space Research, Brazil
Master's Thesis: Pyrgeometer Characterization and Quality Control of Measured Data
Advisor: Dr. José Celso Thomaz

2007-2011 **BSc in Physics** - Federal University of Rio Grande do Norte, Brazil
Undergraduate Project: Ionospheric Plasma Irregularities - Effect on Satellite Telecommunication
Advisor: Prof. Dr. Enivaldo Bonelli

2004-2007 **Soldier** - Brazilian Army, Brazil

Skills

- **Soft Skills**

  Creativity, Proactive, Communicative, Alternative thinking, Problem solving, Teamwork, Adaptability, esults-oriented, Work ethic and experience applying SCRUM

- **Languages**

  Portuguese (native), English, Dutch (on the route to B1)

- **Remote sensing**

  Good experience working with **cloud radars**: **Meteor 50DX (X-band)** from Selex ES, **MIRA 35 (Ka-Band)** from Meteorologische Messtechnik, **FMCW 94 (W-band)** from Radiometer Physics. Good experience working **wind lidar**: **WindCube-200s** from Vaisala (level 1 certification from Vaisala).

- **Radar Forward Operators**

  Introductory experience working with **PAMTRA** and **PyTMatrix**.

- **Atmospheric and Oceanic Sensors**

  Strong knowledge of calibration and measurement technique of **longwave radiation sensors**; experience with measurement procedure using **Expendable Current Profiler(XCP)**, **UnderwayCTD**, **Radiosonde**.

- **Data Acquisition**

  Knowledge programming **datalogger CR3000** from Campbell Scientific.

- **Coding and Development**

  Good knowledge of scientific programming; all projects were developed in **Python** and **Shell-Script**. Introductory knowledge developing in **C**. Good experience automatazing data pipelies using **Apache Airflow** and knowledge of **Docker**

- **Version Control**

  Good experience with version control system (**Git**) and repository (**Bitbucket**, **GitHub**, **GitLab**).

- **Operational Systems**

  Good experience working with **Linux/Unix-based** and **Mac OS X** operating systems.

- **Prototyping Platform**

  Good experience working with **Arduino** and **Raspberry Pi** open-source prototyping platforms.

Experimental Campaigns

6 May 2026 - 10 May 2026 TurPhyCloud
Radar Expert

Technical visit and remote sensing instrument deployment **planning for the TurPhyCloud experiment** - Utö, Finland

13 Sep 2021 - 3 Oct 2021 CMTRACE
Assistant

Planning of the experimental setup and the development of the data processing package of the **Tracing Convective Momentum Transport in Complex Cloudy Atmospheres Experiment** - Cabauw, Netherlands

1 Nov 2018 - 21 Feb 2019 Tripex-Pol
Assistant

Assistence during the **Triple-frequency and Polarimetric Radar Experiment** - Jülich, Germany

20 Nov 2013 - 25 Nov 2013 São Pedro and São Paulo Rocks Research Program
Scientist crew

Scientific cruise participation on board of the Brazilian Navy Araguari Ocean Patrol Vessel for **maintenance and data retrieval from meteorological stations in São Pedro and São Paulo rocks**.

04 Jun 2013 - 24 Jun 2013 Prediction and Research Moored Array in the Atlantic (PIRATA)
Scientist crew

Scientific cruise participation on board of the Ocean Stalwart ship for **maintenance of oceanic buoys and measuring oceanic/atmospheric variables using XCP, underwayCTD, and radiosonde**.

15 Feb 2013 - 25 Mar 2013 Prediction and Research Moored Array in the Atlantic (PIRATA)
Scientist crew

Scientific cruise participation on board of the Ocean Stalwart ship for **maintenance of oceanic buoys and measuring oceanic/atmospheric variables using XCP, underwayCTD, and radiosonde**.

Prize

2019 - 39th International Conference on Radar Meteorology (Japan)
Talk: Investigating snow aggregation close to the melting layer using novel ground-based triple-frequency observations (**2nd best talk**)

Dataset

Dias Neto, J. (2022). **The Tracing Convective Momentum Transport in Complex Cloudy Atmospheres Experiment - Level 2**. *Zenodo.* https://doi.org/10.5281/zenodo.6926605

Dias Neto, J. (2022). **The Tracing Convective Momentum Transport in Complex Cloudy Atmospheres Experiment - Level 1**. *Zenodo.* https://doi.org/10.5281/zenodo.6926483

Dias Neto, J., Kneifel, S., Ori, D. (2019). **The TRIple-frequency and Polarimetric radar Experiment for improving process observation of winter precipitation (version 2)**, *Zenodo*, http://doi.org/10.5281/zenodo.1341390

Research Python Packages

Dias Neto, J., Castelão, G., (2023). **lidarwind: A Python package for retrieving wind profiles from Doppler lidar observations**, *GitHub:* https://github.com/jdiasn/lidarwind
Zenodo: https://doi.org/10.5281/zenodo.7026548

Dias Neto, J., Castelão, G., (2020). **McRdar: an Open Source Python package to simulate the multi-frequency radar variables using the output from McSnow**, *Zenodo*, https://doi.org/10.5281/zenodo.3723886

Publications

Dias Neto, J., Castelão, G. P., (2022) **lidarwind: A Python package for retrieving wind profiles from Doppler lidar observations**, *Journal of Open Source Software*, https://doi.org/10.21105/joss.04852

Dias Neto, J., Nuijens, L., Unal C., Knoop, S., (2022) **Combined wind lidar and cloud radar for high-resolution wind profiling**, *Earth System Science Data*, https://doi.org/10.5194/essd-15-769-2023

von Terzi, L., **Dias Neto, J.**, Ori, O., Myagkov, A., Kneifel, S., (2022). **Ice microphysical processes in the dendritic growth layer: A statistical analysis combining multi-frequency and polarimetric Doppler cloud radar observations** *Atmospheric Chemistry and Physics* , https://doi.org/10.5194/acp-2022-263

Karrer, M., **Dias Neto, J.**, von Terzi, L., Kneifel, S., (2022). **Melting Behavior of Rimed and Unrimed Snowflakes Investigated With Statistics of Triple-Frequency Doppler Radar Observations**, *JGR Atmospheres*, https://doi.org/10.1029/2021JD035907

Ori, D., Schemann, V., Karrer, M., **Neto, J.D.**, von Terzi, L., Seifert, A. and Kneifel, S. (2020). **Evaluation of ice particle growth in ICON using statistics of multi-frequency Doppler cloud radar observations**, *Q J R Meteorol Soc.*, https://doi.org/10.1002/qj.3875.

Mróz, K., Battaglia, A., Kneifel, S., D'Adderio, L. P.,**Dias Neto, J.**, (2020). **Triple-frequency Doppler retrieval of characteristic raindrop size**, *Earth and Space Science*, https://doi.org/10.1029/2019EA000789.

Dias Neto, J., Kneifel, S., Ori, D., Trömel, S., Handwerker, J., Bohn, B., Hermes, N., Mühlbauer, K., Lenefer, M., and Simmer, C,(2019). **The TRIple-frequency and Polarimetric radar Experiment for improving process observations of winter precipitation**, *Earth System Science Data*, 2019, 11, pp.845-863. https://doi.org/10.5194/essd-11-845-2019.

Kneifel, S., **Dias Neto, J.**, Ori, D., Moisseev, D., Tyynelä, J., Adams, I. S., Kuo, Kwo-Sen, Bennartz, R., Berne, A., Clothiaux, E. E., Eriksson, P., Geer, A. J., Honeyager, R., Leinonen, J., Westbrook, C. D., (2018). **Summer Snowfall Workshop: Scattering Properties of Realistic Frozen Hydrometeors from Simulations and Observations, as well as Defining a New Standard for Scattering Databases**, *Bulletin of the American Meteorological Society*, 2018, 99, pp.55-55. https://doi.org/10.1175/BAMS-D-17-0208.1.

Dias Neto, J., Thomaz Júnior, J. C., and Urbano Neto, D. F. (2016). **Mathematical Adjustment Method for Validation of Longwave Radiation Sensor Measurements**, *Revista Brasileira de Meteorologia*, 2016, vol.31, n.1, pp.37-44. http://dx.doi.org/10.1590/0102-778620140022.